NTRK1 fusion-hotspot benchmark report

Study: thca_tcga_pan_can_atlas_2018

Abstract

This report analyzes NTRK1 gene fusions from the thca_tcga_pan_can_atlas_2018 study, covering 6 fusion events. 6/6 fusions with known frame status (100.0%) were in-frame. 6/6 fusions (100.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=1, not statistically significant at alpha=0.05). No literature benchmark is configured for NTRK1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 6 analyzed fusion events, identifying 5 distinct fusion partner genes. The most frequent partner was IRF2BP2, observed in 2 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 6/6 (100.0%) of in-frame fusions retained the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.156843.

Domain disruption

Domain-disruption analysis was skipped: NTRK1 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

83% of 6 fusions with this gene as the 3' partner join at exon 10, against 1% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.000999001, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection could not determine a breakpoint boundary for this run: all events share a single outcome class; no separation is possible

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=6) and "not_retained" (n=0). For Frame_status, retained: 6/6 (100.0%, 95% CI 61.0%-100.0%); not_retained: 0/0 (unavailable, 95% CI unavailable-unavailable). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, p=unavailable.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: NTRK1: fusion-positive-vs-negative expression comparison skipped; no sample had both established SV assay coverage and measured expression, so no fusion-negative group could be defined.

Genomic position recurrence

Genomic position recurrence produced results for this run (2 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: NTRK1 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: NTRK1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: all events share a single outcome class; no separation is possible

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
IRF2BP2	2	0.3333333333333333	0.08045344250700849	None	None	0.6096657120978346	confidence_certainty, domain_retention, recurrence	0.2785845105177471	1
SQSTM1	1	0.16666666666666666	0.08045344250700849	None	None	0.6096657120978346	confidence_certainty, domain_retention, recurrence	0.20166143359467015	2
SSBP2	1	0.16666666666666666	0.08045344250700849	None	None	0.6096657120978346	confidence_certainty, domain_retention, recurrence	0.20166143359467015	3
TFG	1	0.16666666666666666	0.08045344250700849	None	None	0.6096657120978346	confidence_certainty, domain_retention, recurrence	0.20166143359467015	4
TPM3	1	0.16666666666666666	0.08045344250700849	None	None	0.6096657120978346	confidence_certainty, domain_retention, recurrence	0.20166143359467015	5

domain_retention tables

frame_domain_contingency_table

6	0
0	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
THPA	[6, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	6	6	0	0	None

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
13	6	6	1.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
IRF2BP2	2	2	2	2
SQSTM1	1	1	1	1
SSBP2	1	1	1	1
TFG	1	1	1	1
TPM3	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
1	156874000	156875000	5	EVT-TCGA-DJ-A4UP-01-1, EVT-TCGA-DJ-A4UQ-01-2, EVT-TCGA-EL-A3D4-01-6, EVT-TCGA-EM-A3AO-01-5, EVT-TCGA-ET-A40S-01-3
1	156875000	156876000	1	EVT-TCGA-CE-A3MD-01-4

exact_position_recurrence

chromosome	position	n_events	event_ids
1	156874571	5	EVT-TCGA-DJ-A4UP-01-1, EVT-TCGA-DJ-A4UQ-01-2, EVT-TCGA-EL-A3D4-01-6, EVT-TCGA-EM-A3AO-01-5, EVT-TCGA-ET-A40S-01-3

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
399	5	1	0	False

junction_concentration tables

junction_counts

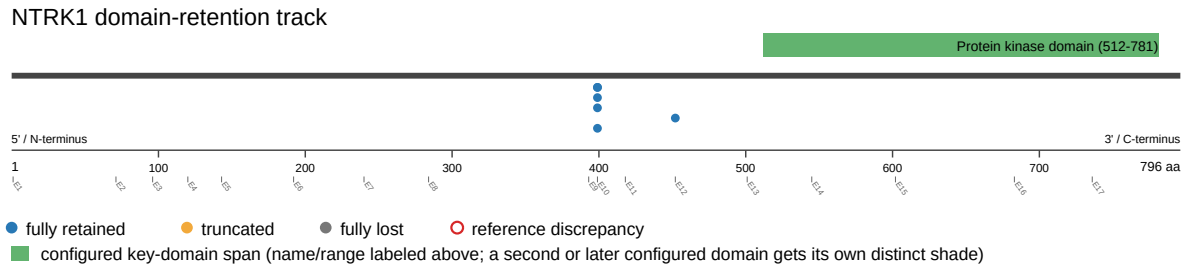
role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	10	5	6	0.8333333333333334	0.010892091185117355	9.040252385738047
three_prime	12	1	6	0.16666666666666666	0.03171237168056292	0.7549515362259704

Per-event results (results.tsv)

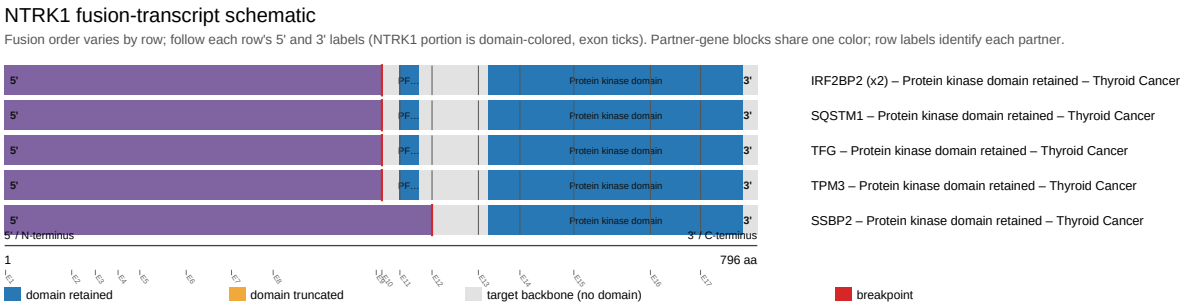
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-DJ-A4UP-01-1	TCGA-DJ-A4UP-01	TCGA-DJ-A4UP	Thyroid Cancer	THPA	IRF2BP2-NTRK1	IRF2BP2	in-frame	three_prime	156874571	10	399	False	399	True	exact match	retained	1.0	False												Protein kinase domain			NA | IRF2BP2-NTRK1 Fusion	in-frame
EVT-TCGA-DJ-A4UQ-01-2	TCGA-DJ-A4UQ-01	TCGA-DJ-A4UQ	Thyroid Cancer	THPA	IRF2BP2-NTRK1	IRF2BP2	in-frame	three_prime	156874571	10	399	False	399	True	exact match	retained	1.0	False												Protein kinase domain			NA | IRF2BP2-NTRK1 Fusion	in-frame
EVT-TCGA-ET-A40S-01-3	TCGA-ET-A40S-01	TCGA-ET-A40S	Thyroid Cancer	THPA	SQSTM1-NTRK1	SQSTM1	in-frame	three_prime	156874571	10	399	False	399	True	exact match	retained	1.0	False												Protein kinase domain			NA | SQSTM1-NTRK1 Fusion	in-frame
EVT-TCGA-CE-A3MD-01-4	TCGA-CE-A3MD-01	TCGA-CE-A3MD	Thyroid Cancer	THPA	SSBP2-NTRK1	SSBP2	in-frame	three_prime	156875520	12	452	False	452	True	exact match	retained	1.0	False												Protein kinase domain			NA | SSBP2-NTRK1 Fusion	in-frame
EVT-TCGA-EM-A3AO-01-5	TCGA-EM-A3AO-01	TCGA-EM-A3AO	Thyroid Cancer	THPA	TFG-NTRK1	TFG	in-frame	three_prime	156874571	10	399	False	399	True	exact match	retained	1.0	False												Protein kinase domain			NA | TFG-NTRK1 Fusion	in-frame
EVT-TCGA-EL-A3D4-01-6	TCGA-EL-A3D4-01	TCGA-EL-A3D4	Thyroid Cancer	THPA	TPM3-NTRK1	TPM3	in-frame	three_prime	156874571	10	399	False	399	True	exact match	retained	1.0	False												Protein kinase domain			NA | TPM3-NTRK1 Fusion	in-frame

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.